

Multiple Choice (10 marks)

1. Light of wavelength 400 nm is incident upon lithium ($\phi = 2.93$ eV). Calculate the stopping potential in V.
 - (a) 0.87
 - (b) -0.23
 - (c) 0.07
 - (d) 0.57
 - (e) 0.97
2. A voltage will be induced in a wire loop when the magnetic field within that loop
 - (a) is amplified
 - (b) converts to magnetic energy
 - (c) is parallel to the electric field
 - (d) changes
 - (e) is at right angles to the electric field
3. Calculate the binding energy of the hydrogen atom in its ground state.
 - (a) 14.1 eV
 - (b) 10.6 eV
 - (c) 11.2 eV
 - (d) 17.0 eV
 - (e) 15.3 eV
4. B^{10} is bombarded with neutrons, and α particles are observed to be emitted. What is the residual nucleus?
 - (a) ${}^8_4\text{Be}$
 - (a) ${}^{13}_6\text{C}$
 - (a) ${}^7_4\text{Be}$
 - (a) ${}^6_3\text{Li}$
 - (a) ${}^5_2\text{He}$
5. The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons

- (a) an average of 10 times, with an rms deviation of about 5
- (b) an average of 10 times, with an rms deviation of about 3
- (c) an average of 10 times, with an rms deviation of about 0.1
- (d) an average of 10 times, with an rms deviation of about 1
- (e) an average of 10 times, with an rms deviation of about 6

True / False (10 marks)

Answer True or False

- 6. True or False: The resistance of an electric toaster operating at 120 volts and drawing 5 amperes is 28 ohms.
- 7. True or False: In a spring-block oscillator, the maximum speed of the block is inversely proportional to the amplitude of oscillation.
- 8. True or False: The combination of a concave lens and a convex lens results in a positive refracting power of 5 diopters.
- 9. True or False: A step-up transformer can increase both current and power simultaneously in an electrical circuit.
- 10. True or False: An object subjected to a force of 2000 dynes and accelerating at 500 cm/s^2 has a mass of 1 gram.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the relationship between capacitance, potential difference, and charge in capacitors. Given a capacitance of 2 mF and a potential difference of 5 V, calculate the charge on the positive plate and analyze the implications of your result.
- 12. Discuss the conditions under which there are Avogadro's number of translational states available for O_2 in a volume of 1000 cm^3 , particularly focusing on temperature.

End of Paper